

Evaluating pain tolerance when using turtle for class assignments

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1 Abstract

This paper discusses and analyses potential users' pain tolerance when using the well-known and well-feared Python library turtle, which must be handled with extreme care to prevent slow code execution, getting a poor mark from the teacher, spiralling into depression and existential dread, and the total collapse of civilisation itself.¹ The conclusion is that turtle is pretty damn terrible.²

2 Introduction

Python is a high-level programming language and one of the most popular programming languages Out There³ due to the sheer amount of random stuff you can do with it (or force it to do). However, this handy tool has been manipulated for the creation of evil (most things are, unfortunately) in the form of the library turtle.

The main function of this ~~abomination~~ Python library is to draw things. Within the domain of high school coding classes, a particular favorite of the dozens of tedious tasks teachers like to assign is the dreaded 'draw this using turtle'.⁴ Common shapes requested by the ~~Dark Lords~~ teachers are straight lines, squares, triangles, circles, stars, and very loosely accurate recreations of Flinders Street Station⁵, in that order.

¹I debated here whether this was too dramatic, but alas, I must give an accurate description. This is real life, you know.

²Hey, you knew it was coming.

³'Out There' referring to my laptop, Newcastle, Rivendell, Michigan, hyperspace, Antarctica, high schools, Wean and Doherty, Venus, Dragonstone, Hogwarts, programmer asylums, Black Mesa, waffles, Libraria, and the multiverse, among other places.

⁴*cue screams of terror*

⁵This is a real task. We were given one lesson. One. Lesson.

3 Examples of Turtle use cases

As aforementioned, common Turtle use cases are drawing straight lines, squares, triangles, circles, stars, and very loosely accurate recreations of Flinders Street Station. An example is provided below of the turtle code required to draw a line of length 100.⁶

```
import turtle
t = turtle.Turtle()
t.forward(100)
```

Here is a snippet from the start of a student's turtle code required to draw a very loosely accurate recreation of Flinders Street Station. The rest was omitted due to the dangers of providing you, the reader, with code that will undoubtedly ruin your life as you know it. I mean, just look at this disaster.

```
from turtle import*

speed(0)
setup(1000, 600)
color("DarkOliveGreen")
pensize(3)

#Playground_Green_Tall
goto(0,-150)
left(90)
forward(325)
```

The whole code is actually around 500 lines long.

Here is a piece of code provided by an unfortunate Stack Overflow user consisting of four lines.

```
import turtle
turtlescreen
turtle.pos(400,400)
turtle.forward()
```

Considering this user managed to cram three errors into four lines of code, it is really easy to imagine *just how many* errors may lie in a five-hundred-line abomination, lurking in wait, to the screams of once-hopeful programming students everywhere.

⁶My third attempt to write working turtle code crashed my computer. I have thus resorted to using an online turtle sandbox.

4 ~~Structured kidnappings~~ Interviews Surveys

In the interest of finding fellow programmers who have suffered extensive trauma⁷ at the hands of Python's turtle, I asked some ~~unwilling test subjects~~ friends who had been in the infamous forced-to-use-turtle-for-a-class-project class to fill out a form regarding usage of turtle. This experiment was conducted for the purpose of determining where the agony of using turtle becomes too great to continue coding and breaks a coder's will so thoroughly they resort to - insert gasp of shock - learning C++. (Or not.)

Along with the survey, I walked⁸ around school and asked people what they thought of turtle. Responses included 'worst programming experience of my life', 'just writing the same line over and over again', and equally correct expressions. In fact, after consulting one of the saner programming teachers at school (of which there are not many) I was able to garner that the coding department is heavily considering not teaching turtle. See, after a while, even the very beings that forced this terror upon us all are realizing the sheer pain their actions inflict!

5 Results

Warning: This section contains examples of high school humor, which may be terrible.

Along with the results of the survey came some very enlightening reviews, including:

- Met a turtle, killed my dog and reported me to the IRS for tax fraud, would not use again.
- used a turtle, he ended up killing my neighbour and is now on the run evading taxes, highly don't recommend. 1 star review.⁹ Absolute garbage
- I have been arrested by the FBI after I imported turtle and it drew top secret government documents in a window. Would not recommend.

Had to include those first. For the actual results of the survey, 8 out of 10 people opted to describe turtle as an abomination of nature (as it clearly is), while 2 out of 10 picked the option 'would rather die than use it again'. Given that both responses were defined as equally bad, it can be inferred that the usage of turtle will lead to one-fifth of programmers dying due to sheer fear and depression. This is a rather compelling point from which it is safe to assume that eradicating turtle entirely will drastically lower high school programmer mortality rates. A supplementary question, 'do you hate turtle?', yielded 100% truthy values. In addition, not a single person out of the n=10 people surveyed (n=10 because I am a programmer, and thus have a limited selection of friends) gained any useful skills through the usage of turtle.

⁷which, notably, gives the affected victim the tendency to use a lot of footnotes

⁸gasp. extended gasp. audible gasp. Seriously though, DFS isn't very efficient. I should have enlisted multiple victims and sent everyone out at the same time.

⁹Really? Generous, you are.

100% of people surveyed would not draw a line, square, circle or doubtfully accurate Flinders Street Station. 1 person opted to draw a radioactivity sign (ok, it is pretty cool, but seriously, why??) and a triangle. From this it is evident that 1 out of 10 programmers using turtle are insane, as no sane person would willingly use turtle. Thus, turtle promotes insanity, chaos, and disorder of the natural state of the universe.

During analysis, I found it necessary to query further to investigate just how insane turtle makes people. Bear in mind that the accuracy of this entire paper may be affected by the very minor case of serious brain damage I have received from my usage of turtle at a young age. Responses received to the question ‘what have you gotten out of turtle?’ included: a deep sense of sadness, a broken arm, breadcrumbs, pigeons, trust issues, problems with socializing, less friends, and avocados. From this data it is now evident that the previous statement is misinformed and everyone who uses turtle is insane.

6 Conclusion

As previously stated, turtle is Pretty Damn Terrible. If you, the reader, are considering using it, may I politely ask *what’s wrong with you!?* If you have been assigned a turtle task in class¹⁰, I suggest any one of the following excuses: cannot install, dog ate the homework, turtle ate the homework, laptop broke, will to live broke, etc, etc.

7 Related work

There are undoubtedly countless essays out there about the sheerly moronic idea of using turtle, but I, in all my glorious laziness¹¹ did not bother to look for them.

8 References

<https://stackoverflow.com/questions/60050020/turtle-library-in-python>¹²

The start of a friend’s turtle code to draw Flinders Street Station.

The Wikipedia page for Python. Oh, you want the link, do you? Ha ha, very funny. Just kidding, here it is: [https://en.wikipedia.org/wiki/Python_\(programming_language\)](https://en.wikipedia.org/wiki/Python_(programming_language))

The eyewitness and testimony accounts of several like-minded tortured souls.

¹⁰scream loudly to express the unfathomable pain that courses within your soul

¹¹Nah, just spending way too much time writing footnotes and slowly, ever so slowly, digging through broken code. Dang.

¹²I really do feel sorry for them. 3 errors in 4 lines of code, and that’s just the start of it all...